Chapter 21: G-Methods for Time-Varying Treatments

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Chapter 20 showed that, with treatment-confounder feedback, traditional adjustment methods give a non-null estimate even when a time-varying treatment has no effect at all. This chapter shows that the **g-methods** – the g-formula, IP weighting, g-estimation, and doubly robust versions that combine them – recover the correct (null) answer in the same dataset. Throughout, we compare **static** strategies under the identifiability conditions of Chapter 19: sequential exchangeability, positivity, and consistency.

This chapter is based on Hernán and Robins (2020, chap. 21, pp. 277-303).

Roadmap: each method was introduced earlier for a time-fixed treatment: IP weighting of marginal structural models (Chapter 12), the g-formula (Chapter 13), and g-estimation of structural nested models (Chapter 14) (the book's introduction to Chapter 21 says Chapter 15 (Hernán and Robins 2020, 277), but g-estimation is the subject of Chapter 14). All three g-methods rest on the same identifying assumptions but model different parts of the data distribution: the g-formula models the outcome and covariates, IP weighting models treatment, and g-estimation models treatment plus a structural model for the effect of each "blip" of treatment. Section 21.5 adds censoring, and Section 21.6 relates the g-formula to other identifying formulas.

1 21.1 The G-Formula for Time-Varying Treatments (pp. 277-281)

The running example is the sequentially randomized experiment of Chapter 20 (Table 20.1, reproduced in the book as Table 21.1), with treatments A_0, A_1, a confounder L_1 measured between them, and an outcome Y. There is no L_0 in this study.

Table 1: The sequentially randomized experiment of Table 21.1 (Hernán and Robins 2020, 277); 32,000 individuals in total.

N	A_0	L_1	A_1	Mean Y
2400	0	0	0	84
1600	0	0	1	84
2400	0	1	0	52
9600	0	1	1	52
4800	1	0	0	76
3200	1	0	1	76
1600	1	1	0	44
6400	1	1	1	44

For the effect of the time-fixed treatment A_1 alone, the g-formula is ordinary standardization: $\mathrm{E}[Y^{a_1}] = \sum_{l_1} \mathrm{E}[Y \mid A_1 = a_1, L_1 = l_1] \Pr[L_1 = l_1]$. For the joint treatment (A_0, A_1), the weights become the distribution of L_1 **given past treatment**.

Definition 1.1 (The G-Formula for Time-Varying Treatments)**.** For two time points, the g-formula for the static strategy (a_0, a_1) is

$$\sum_{l_1} \mathrm{E}[Y \mid A_0 = a_0, A_1 = a_1, L_1 = l_1]\, f(l_1 \mid a_0).$$

For $K+1$ time points $k = 0, \ldots, K$ and strategy $\bar{a}$, it is

$$\sum_{\bar{l}} \mathrm{E}\big[Y \mid \bar{A} = \bar{a}, \bar{L} = \bar{l}\big] \prod_{k=0}^{K} f\big(l_k \mid \bar{a}_{k-1}, \bar{l}_{k-1}\big),$$

where the sum is over all covariate histories $\bar{l}$. Under sequential exchangeability for $Y^{\bar{a}}$ given $(\bar{L}_k, \bar{A}_{k-1})$ at each k, positivity, and consistency, the g-formula equals $\mathrm{E}[Y^{\bar{a}}]$.

Robins (1986, 1987) introduced the g-formula for time-varying treatments, as cited in Hernán and Robins (2020, 278). Every factor is conditional on prior treatment and covariate history; this conditioning is unnecessary in the time-fixed case, where treatment and confounders are measured at a single time.

⚠ Positivity Is Needed to Compute the G-Formula

Positivity: the g-formula is computable only if, for every l_1 with $f(l_1 \mid a_0) \neq 0$, some individuals have $(A_0 = a_0, A_1 = a_1, L_1 = l_1)$ (the definition of positivity in Technical Point 19.2).

Fine Point 21.1: Treatment and Covariate History

The relevant history at time k is the set of treatments and confounders needed for conditional exchangeability of A_k. Usually this is the chronological past, but confounders can in principle lie in the temporal future of treatment (Fine Point 7.4), and adjusting for some past variables can create M-bias (Figure 7.4) (Hernán and Robins 2020, 280).

Example 1.1 (G-Formula in the Sequentially Randomized Experiment)**. Never treat**, $(a_0, a_1) = (0, 0)$. Among the 16,000 individuals with $A_0 = 0$, $2400 + 1600 = 4000$ have $L_1 = 0$ and $2400 + 9600 = 12000$ have $L_1 = 1$, so $f(L_1 = 0 \mid A_0 = 0) = 0.25$ and $f(L_1 = 1 \mid A_0 = 0) = 0.75$. Then

$$\begin{aligned} \mathrm{E}[Y^{a_0=0,a_1=0}] &= \mathrm{E}[Y \mid 0, 0, L_1 = 0] \times 0.25 + \mathrm{E}[Y \mid 0, 0, L_1 = 1] \times 0.75 \\ &= 84 \times 0.25 + 52 \times 0.75 \\ &= 21 + 39 = 60. \end{aligned}$$

Always treat, $(a_0, a_1) = (1, 1)$. Among the 16,000 individuals with $A_0 = 1$, $4800 + 3200 = 8000$ have $L_1 = 0$, so $f(L_1 = 0 \mid A_0 = 1) = 0.5$, and

$$\begin{aligned} \mathrm{E}[Y^{a_0=1,a_1=1}] &= 76 \times 0.5 + 44 \times 0.5 \\ &= 38 + 22 = 60. \end{aligned}$$

The g-formula estimate of $\mathrm{E}[Y^{1,1}] - \mathrm{E}[Y^{0,0}]$ is $60 - 60 = 0$, the correct null value (Hernán and Robins 2020, 278).

1.1 The G-Formula as a Simulation

Remark 1.1 (The g-formula simulates the counterfactual joint distribution). Under sequential exchangeability for Y and $\bar{L}$ jointly, the g-formula simulates the joint distribution of the counterfactuals $(Y^{\bar{a}}, \bar{L}^{\bar{a}})$ that would have been observed had everybody followed strategy $\bar{a}$. On the tree graph (Figures 21.1-21.2), this means building a new tree in which everyone receives a_0 and a_1 with probability 1, while $\Pr[L_1 = l_1 \mid A_0 = a_0]$ and $\mathrm{E}[Y \mid A_0 = a_0, A_1 = a_1, L_1 = l_1]$ keep their observed values.

Two Cautions About the G-Formula

Two cautions (Hernán and Robins 2020, 279):

1. **The value of the g-formula depends on what is in L.** If we wrongly omit L_1 (believing it is not a confounder), the g-formula reduces to $\mathrm{E}[Y \mid A_0 = a_0, A_1 = a_1]$, which has no causal interpretation when L_1 affects A_1 (Figure 20.8).
2. **The g-formula can be causal even when its components are not.** Under Figure 20.9, where only static sequential exchangeability holds, the g-formula including L_1 still identifies $\mathrm{E}[Y^{\bar{a}}]$, yet neither $\Pr[L_1 = l_1 \mid A_0 = a_0]$ nor $\mathrm{E}[Y \mid A_0 = a_0, A_1 = a_1, L_1 = l_1]$ equals its counterfactual analogue. In a sequentially randomized trial (Figures 20.1 and 20.2) the components are causal as well: $\Pr[L_1 = l_1 \mid A_0 = a_0] = \Pr[L_1^{a_0} = l_1]$ and $\mathrm{E}[Y \mid A_0 = a_0, A_1 = a_1, L_1 = l_1] = \mathrm{E}[Y^{a_0,a_1} \mid L_1^{a_0} = l_1]$, so the g-formula is $\sum_{l_1} \mathrm{E}[Y^{a_0,a_1} \mid L_1^{a_0} = l_1] \Pr[L_1^{a_0} = l_1] = \mathrm{E}[Y^{a_0,a_1}]$.

1.2 Estimation: the Plug-In (Parametric) G-Formula

Definition 1.2 (Plug-in and parametric g-formula)**.** With many confounders or time points, the components must be estimated, for example

- a linear regression model for $\mathrm{E}\big[Y \mid \bar{A} = \bar{a}, \bar{L} = \bar{l}\big]$, and
- logistic regression models for the discrete confounders L_k, $k \neq 0$ (the distribution of L_0 can be estimated without a model, as in Section 13.3).

Plugging the estimates into the formula gives the **plug-in g-formula**; when the estimates come from parametric models, the **parametric g-formula**.

Definition 1.3 (The g-formula for a random strategy)**. Random strategies**: under sequential exchangeability the g-formula also handles a random strategy f^{int}, such as "at each k, independently treat with probability 0.3" (so $f^{int}(1 \mid \bar{a}_{k-1}, \bar{l}_k) = 0.3$). The general expression is

$$\sum_{\bar{a},\bar{l}} \mathrm{E}\big[Y \mid \bar{A} = \bar{a}, \bar{L} = \bar{l}\big] \prod_{k=0}^{K} f(l_k \mid \bar{a}_{k-1}, \bar{l}_{k-1}) \prod_{k=0}^{K} f^{int}(a_k \mid \bar{a}_{k-1}, \bar{l}_k).$$

Replacing f^{int} by the observed $f(a_k \mid \bar{a}_{k-1}, \bar{l}_k)$ gives the observed mean of Y. For a deterministic strategy, f^{int} is 1 for the mandated treatment values and 0 otherwise, so the f^{int} factors and the sum over $\bar{a}$ can be dropped.

Technical Point 21.1: The g-formula Density

For a static strategy $\bar{a}$, the g-formula density of $(Y, \bar{L})$ at $(y, \bar{l})$ is $f(y \mid \bar{a}_K, \bar{l}_K) \prod_{k=0}^{K} f(l_k \mid \bar{a}_{k-1}, \bar{l}_{k-1})$, and integrating out $\bar{l}$ gives the g-formula density of Y. For a deterministic dynamic strategy g, replace each a_k by the value a_k^g that g assigns given the simulated history.

Software for the Parametric G-Formula

Software: the `gfoRmula` R package (Lin et al. 2019) is on CRAN, and a GFORMULA SAS macro is available on GitHub (Hernán and Robins 2020, 281).

2 21.2 IP Weighting for Time-Varying Treatments (pp. 282-285)

For the time-fixed treatment A_1 alone (Chapter 12), the weights are $W^{A_1} = 1/f(A_1 \mid L_1)$ or $SW^{A_1} = f(A_1)/f(A_1 \mid L_1)$. For a time-varying treatment, the denominator becomes each individual's probability of the **treatment history** they received, given their treatment and covariate history.

Definition 2.1 (IP Weights for Time-Varying Treatments)**.** With $A_{-1} \equiv 0$ by definition, the **nonstabilized** IP weights are

$$W^{\bar{A}} = \prod_{k=0}^{K} \frac{1}{f(A_k \mid \bar{A}_{k-1}, \bar{L}_k)},$$

and the **stabilized** IP weights are

$$SW^{\bar{A}} = \prod_{k=0}^{K} \frac{f(A_k \mid \bar{A}_{k-1})}{f(A_k \mid \bar{A}_{k-1}, \bar{L}_k)}.$$

For $K = 1$: $W^{\bar{A}} = \dfrac{1}{f(A_0 \mid L_0)} \times \dfrac{1}{f(A_1 \mid A_0, L_0, L_1)}$.

Proposition 2.1 (IP weighting identifies the counterfactual mean)**.** *Under the identifiability conditions (sequential exchangeability, positivity, and consistency),* $\mathrm{E}[Y^{a_0,a_1}]$ *equals the mean* $\mathrm{E}_{ps}[Y \mid A_0 = a_0, A_1 = a_1]$ *in the pseudo-population created by either set of weights in Definition 2.1.*

As in Technical Point 12.2, the pseudo-population mean equals $\mathrm{E}\left[W^{\bar{A}} Y I(A_0 = a_0, A_1 = a_1)\right]$ (nonstabilized) or the ratio $\mathrm{E}\left[SW^{\bar{A}} Y I(\cdot)\right] / \mathrm{E}\left[SW^{\bar{A}} I(\cdot)\right]$ (stabilized), whether or not sequential exchangeability holds (Hernán and Robins 2020, 282).

In the pseudo-population, the probability of treatment at each k is the constant 1/2 (nonstabilized weights) or depends at most on past treatment (stabilized weights), so sequential *unconditional* exchangeability holds there, and $\mathrm{E}[Y^{\bar{a}}] - \mathrm{E}\left[Y^{\bar{a}'}\right] = \mathrm{E}_{ps}\left[Y \mid \bar{A} = \bar{a}\right] - \mathrm{E}_{ps}\left[Y \mid \bar{A} = \bar{a}'\right]$. In a true sequentially randomized trial the treatment probabilities are known by design, so nonstabilized-weight estimates are unbiased; in observational studies they must be estimated, for example by a pooled logistic model for $\Pr[A_k = 1 \mid \bar{A}_{k-1}, \bar{L}_k]$ that includes functions of time k.

⚠ The Denominator Model Must Be Correct

A misspecified denominator model biases the estimate; a misspecified numerator model for $f(A_k \mid \bar{A}_{k-1})$ does not.

Example 2.1 (IP Weighting in the Sequentially Randomized Experiment)**.** There is no L_0, so the denominator is $f(A_0)f(A_1 \mid A_0, L_1)$, with $f(A_0 = 0) = 16000/32000 = 0.5$. For the never-treated rows of Table 1:

- $(A_0, L_1, A_1) = (0, 0, 0)$: $f(A_1 = 0 \mid A_0 = 0, L_1 = 0) = 2400/4000 = 0.6$, so $W^{\bar{A}} = 1/(0.5 \times 0.6) = 10/3$ and the row contributes $2400 \times 10/3 = 8000$ pseudo-individuals.
- $(0, 1, 0)$: $f(A_1 = 0 \mid A_0 = 0, L_1 = 1) = 2400/12000 = 0.2$, so $W^{\bar{A}} = 1/(0.5 \times 0.2) = 10$ and the row contributes $2400 \times 10 = 24000$.

The 32,000 pseudo-individuals with $(A_0, A_1) = (0, 0)$ give

$$\begin{aligned}\mathrm{E}_{ps}[Y \mid A_0 = 0, A_1 = 0] &= 84 \times \frac{8000}{32000} + 52 \times \frac{24000}{32000} \\ &= 21 + 39 = 60.\end{aligned}$$

The same calculation for $(1, 1)$ also gives 60, so the IP weighted effect estimate is 0. The whole pseudo-population has $4 \times 32000 = 128000$ individuals, the 32,000 study individuals times the 4 static strategies (Hernán and Robins 2020, 282–83).

Remark 2.1 (Equal nonparametric estimates are not evidence of causality). The nonparametric g-formula and IP weighted estimates are exactly equal. That equality has nothing to do with causality: it holds even when the identifiability conditions fail and neither estimate has a causal interpretation. Stabilized weights give the same estimate of 0 here (check for yourself) (Hernán and Robins 2020, 283).

Estimate the Counterfactual Mean Both Ways

Comparing the two parametric estimators: if the parametric g-formula and the parametric IP weighted estimates differ by more than sampling variability (quantified by bootstrapping the difference), then at least one set of models is misspecified, whether or not the identifiability

assumptions hold. Similar estimates do not prove the models are correct, because both may be biased in the same direction. The book recommends estimating $\mathrm{E}[Y^{\bar{a}}]$ both ways and, if they differ substantially by a prespecified criterion, revisiting the models (Hernán and Robins 2020, 284).

2.1 Marginal Structural Models

Definition 2.2 (Marginal structural mean model for a time-varying treatment)**.** There are far more strategies $\bar{a}$ than can be estimated one at a time, so we pool information with a **marginal structural mean model**, for example

$$\mathrm{E}[Y^{\bar{a}}] = \beta_0 + \beta_1 \,\mathrm{cum}(\bar{a}), \qquad \mathrm{cum}(\bar{a}) = \sum_{k=0}^{K} a_k.$$

This model is unsaturated: one unknown per strategy on the left, two parameters on the right. The average causal effect is $\mathrm{E}[Y^{\bar{a}}] - \mathrm{E}\left[Y^{\bar{a}=\bar{0}}\right] = \beta_1 \times \mathrm{cum}(\bar{a})$.

Proposition 2.2 (IP weighted least squares is consistent for the marginal structural model)**.** *Suppose the marginal structural model of Definition 2.2 is correctly specified, and the identifiability conditions (sequential exchangeability, positivity, and consistency) hold. Let $\hat{\beta}_1$ be the coefficient of* $\mathrm{cum}(\bar{A})$ *in the fit of the model* $\mathrm{E}[Y \mid \bar{A}] = \theta_0 + \theta_1 \,\mathrm{cum}(\bar{A})$ *by weighted least squares with weights $SW^{\bar{A}}$ or $W^{\bar{A}}$ of Definition 2.1 (known, or estimated from a correctly specified treatment model). Then $\hat{\beta}_1$ is consistent for the causal β_1, which in general differs from the associational θ_1 of the unweighted data.*

Variance Estimation and Choice of Weights

Variance: use the nonparametric bootstrap, the analytic variance, or a conservative 95% interval from the robust variance estimator. For a non-saturated model, intervals are typically narrower with $SW^{\bar{A}}$ than with $W^{\bar{A}}$, so $SW^{\bar{A}}$ is preferred (Hernán and Robins 2020, 285).

Checking the Marginal Structural Model

Checking the MSM: fitting

$$\mathrm{E}[Y \mid \bar{A}] = \theta_0 + \theta_1 \mathrm{cum}(\bar{A}) + \theta_2 \mathrm{cum}_{-5}(\bar{A}) + \theta_3 \mathrm{cum}(\bar{A})^2$$

with the same weights, where cum_{-5} is cumulative treatment in the final 5 months, gives a 2-degree-of-freedom Wald test of $\theta_2 = \theta_3 = 0$, a test that the original MSM is correctly specified. In practice one may use other summaries of treatment history and flexible functions such as cubic splines.

Definition 2.3 (Marginal structural model with a baseline effect modifier)**. Effect modification**: for a dichotomous *baseline* variable V in L_0, a marginal structural model with effect modification by V is $\mathrm{E}[Y^{\bar{a}} \mid V] = \beta_0 + \beta_1 \mathrm{cum}(\bar{a}) + \beta_2 V + \beta_3 \mathrm{cum}(\bar{a}) V$. Its parameters are estimated by fitting the associational model $\mathrm{E}[Y \mid \bar{A}, V] = \theta_0 + \theta_1 \mathrm{cum}(\bar{A}) + \theta_2 V + \theta_3 \mathrm{cum}(\bar{A}) V$ by weighted least squares with $W^{\bar{A}}$ or, better, $SW^{\bar{A}}(V) = \prod_{k=0}^{K} f(A_k \mid \bar{A}_{k-1}, V) / f(A_k \mid \bar{A}_{k-1}, \bar{L}_k)$.

Proposition 2.3 (IP weighted least squares is consistent for the effect-modification model)**.** *Suppose the marginal structural model of Definition 2.3 is correctly specified, the identifiability conditions (sequential exchangeability, positivity, and consistency) hold, and V is a baseline variable in L_0. Let $\hat{\beta}_1$ and $\hat{\beta}_3$ be the coefficients of* $\text{cum}(\bar{A})$ *and* $\text{cum}(\bar{A})V$ *in the fit of the associational model of Definition 2.3 by weighted least squares with weights $SW^{\bar{A}}(V)$ of Definition 2.3 or $W^{\bar{A}}$ of Definition 2.1 (known, or estimated from a correctly specified model for treatment in the denominator). Then $\hat{\beta}_1$ and $\hat{\beta}_3$ are consistent for the causal β_1 and β_3.*

⚠ Only Baseline Variables as Effect Modifiers

With treatment-confounder feedback, V may include only baseline variables; if V included any L_k with $k > 0$, the weighted-fit coefficients of $\text{cum}(\bar{A})$ and $\text{cum}(\bar{A})V$ could be non-null even though $\beta_1 = \beta_3 = 0$ under the null (Hernán and Robins 2020, 285).

ℹ Technical Point 21.2: IP Weighting for Dynamic Treatment Strategies

For a deterministic dynamic strategy g, the g-formula equals the mean of Y among pseudo-population members who follow g in the pseudo-population created by **nonstabilized** weights. Stabilized weights cannot be used, because their numerator depends on A.

ℹ Technical Point 21.3: The g-null Paradox

Under the sharp null with treatment-confounder feedback, the g-formula does not depend on $\bar{a}$, but non-saturated parametric models for $\text{E}\left[Y \mid \bar{A}, \bar{L}\right]$ and $f(l_k \mid \cdot)$ with variation-independent parameters cannot all be correctly specified when L_k has a discrete component (Robins and Wasserman 1997). The estimated g-formula may therefore falsely reject the null even in a sequentially randomized experiment. In practice the bias seems small relative to random variability. MSMs and structural nested mean models do not suffer from this paradox: both are correctly specified under the null whatever functional form we choose (Hernán and Robins 2020, 286).

3 21.3 A Doubly Robust Estimator for Time-Varying Treatments (pp. 286-288)

Definition 3.1 (Doubly robust estimator)**.** IP weighting needs a correct model for treatment given confounders; the g-formula needs a correct model for the outcome given treatment and confounders. A **doubly robust** estimator is consistent if *either* model is correct, without our knowing which, giving "two chances to get it right" (Hernán and Robins 2020, 286).

3.1 Review: a Doubly Robust Plug-In Estimator for a Time-Fixed Treatment

Algorithm 3.1 (Doubly robust plug-in estimator for a time-fixed treatment)**.** *For a binary A, binary Y, and many confounders L, estimate* $\text{E}[Y^a]$ *in three steps:*

1. *Fit a treatment model and compute $\hat{f}(a \mid L) = \widehat{\Pr}[A = a \mid L]$.*
2. *Among individuals with $A = a$, fit by maximum likelihood an outcome model that includes $\hat{W}^a = 1/\hat{f}(a \mid L)$ as a covariate, such as $b(a, L; \theta) = \text{expit}(\theta_{a,0} + \theta_{a,1}L + \theta_{a,2}\hat{W}^a)$.*
3. *Average the predictions $b(a, L; \hat{\theta})$ over **all** individuals, treated and untreated.*

The difference of the step-3 averages for $a = 1$ and $a = 0$ is a doubly robust estimator of

$\mathrm{E}[Y^{a=1}] - \mathrm{E}[Y^{a=0}]$.

3.2 Extension to Time-Varying Treatments

Algorithm 3.2 (Doubly robust plug-in estimator for a time-varying treatment)**.** *To estimate* $\mathrm{E}\left[Y^{\bar{a}=\bar{1}}\right]$ *("always treated"):*

1. ***Treatment model****: fit a model* $\pi_k(\bar{L}_k;\alpha)$ *for* $\Pr[A_k = 1 \mid \bar{A}_{k-1} = \bar{1}_{k-1}, \bar{L}_k]$*, pooled over persons and times, using only individuals treated through* $k-1$*. For those treated through* m*, compute the time-varying weights* $\hat{W}^{\bar{1}_m} = \prod_{k=0}^{m} 1/\hat{\pi}_k$*.*
2. ***Sequential outcome models****: for* $m = K, K-1, \ldots, 0$*, among individuals treated through* m*, fit a model* $b_m(\bar{L}_m;\beta_m)$ *that includes* $\hat{W}^{\bar{1}_m}$ *as a covariate. The time-*K *model has dependent variable* Y*; the time-*m *model,* $m < K$*, has as dependent variable the predictions* $\hat{B}_{m+1} = b_{m+1}(\bar{L}_{m+1};\hat{\beta}_{m+1})$ *from the time-*$(m+1)$ *model.*
3. ***Average****: estimate* $\mathrm{E}\left[Y^{\bar{a}=\bar{1}}\right]$ *by the sample average of* $\hat{B}_0$ *over all individuals.*

Repeat with $\bar{a} = \bar{0}$ *for "never treated"; the difference estimates the average causal effect.*

> **Choosing the Sequential Outcome Models**
>
> For a binary Y, the time-m model can be logistic, $\mathrm{expit}(\gamma_m X_m + \varsigma_m \hat{W}^{\bar{1}_m})$ with X_m a function of $\bar{L}_m$: although $\hat{B}_{m+1}$ is not 0 or 1, it lies in $[0,1]$. For a continuous Y, use a linear model (Hernán and Robins 2020, 288).

This estimator is due to Bang and Robins (2005), building on Robins (2000). It is a targeted minimum loss-based estimator (TMLE) in the terminology of van der Laan and coauthors (Hernán and Robins 2020, 287).

Proposition 3.1 (The time-varying doubly robust estimator is $K+2$ robust)**.** *Suppose each sequential outcome model* b_m *of alg. 3.2 is fit by solving the score equations of a generalized linear model with a canonical link (logistic for a binary* Y*, linear for a continuous* Y*). For a binary* Y *and* $m < K$*, the response* $\hat{B}_{m+1}$ *lies in* $[0,1]$ *rather than* $\{0,1\}$*, so the logistic fit is a quasi-likelihood fit. Suppose also that each* b_m *includes the weight* $\hat{W}^{\bar{1}_m}$ *as a covariate. Suppose further that positivity holds for the strategy* $\bar{a} = \bar{1}$*. Then the estimator of alg. 3.2 is consistent for the g-formula functional of Definition 1.1 at* $\bar{a} = \bar{1}$ *if (i) the outcome models* b_m *are correct for all* m*, or (ii) the treatment models* π_k *are correct for all* k*. In fact it is* $K+2$ ***robust****: it is also consistent if, for some* $m \in \{0, \ldots, K-1\}$*, the treatment model is correct for times* 0 *to* m *and the outcome model is correct for times* $m+1$ *to* K *(Hernán and Robins 2020, 288). If, in addition, sequential exchangeability and consistency hold, that functional equals* $\mathrm{E}\left[Y^{\bar{a}=\bar{1}}\right]$*, so the estimator is consistent for the counterfactual mean.*

> **Technical Point 21.4: A K+2 Robust Augmented IP Weighted Estimator**
>
> For $K = 1$, let $\hat{b}_1(L_0, L_1) = \hat{\mathrm{E}}[Y \mid L_0, A_0 = 1, L_1, A_1 = 1]$ and $\hat{b}_0(L_0) = \hat{\mathrm{E}}\left[\hat{b}_1(L_0, L_1) \mid A_0 = 1, L_0\right]$ (the iterated conditional expectation, ICE, plug-in estimator of the g-formula averages $\hat{b}_0$), and let $\hat{\pi}_0, \hat{\pi}_1$ estimate $\Pr[A_0 = 1 \mid L_0]$ and $\Pr[A_1 = 1 \mid L_0, L_1, A_0 = 1]$. The augmented IP weighted estimator of Robins et al. (1994) is the sample average of
>
> $$\hat{U}_{TR} = \hat{b}_0(L_0) + \frac{A_0 A_1}{\hat{\pi}_0 \hat{\pi}_1}\left(Y - \hat{b}_1(L_0, L_1)\right) + \frac{A_0}{\hat{\pi}_0}\left(\hat{b}_1(L_0, L_1) - \hat{b}_0(L_0)\right).$$
>
> It is consistent if (a) $\hat{\pi}_0, \hat{\pi}_1$ are consistent (the augmentation terms then average to 0, leaving the IP weighted estimator), (b) both outcome regressions are consistent (the last two terms

then average to 0, leaving the ICE estimator), or (c) $\hat{b}_1$ and $\hat{\pi}_0$ are consistent (Molina et al. 2017): hence "triply", that is $K+2$, robust. A modification due to Tchetgen Tchetgen (2009) is 2^{K+1} (quadruply) robust. Technical Point 21.5 shows that including the terms $A_0A_1/(\hat{\pi}_0\hat{\pi}_1)$ and $A_0/\hat{\pi}_0$ as covariates in the sequential logistic outcome models makes both correction terms average to exactly 0, so the plug-in estimate stays in $[0,1]$; the estimator in the main text is an instance of this construction.

⚠ Fact-Check: A Term in the Book's First Display of the Estimator

Fact-check note: the book's first display of $\hat{U}_{TR}$ (Hernán and Robins 2020, 288) writes the $\hat{b}_1$ term as $-\{A_0A_1/(\hat{\pi}_0\hat{\pi}_1)-1\}\hat{b}_1$, which disagrees with the rearranged form the same Technical Point gives (the one shown here). Expanding the rearranged form term by term gives

$$\hat{b}_0+\frac{A_0A_1}{\hat{\pi}_0\hat{\pi}_1}\left(Y-\hat{b}_1\right)+\frac{A_0}{\hat{\pi}_0}\left(\hat{b}_1-\hat{b}_0\right)=\frac{A_0A_1Y}{\hat{\pi}_0\hat{\pi}_1}-\left\{\frac{A_0A_1}{\hat{\pi}_0\hat{\pi}_1}-\frac{A_0}{\hat{\pi}_0}\right\}\hat{b}_1-\left\{\frac{A_0}{\hat{\pi}_0}-1\right\}\hat{b}_0,$$

so the "-1" inside the braces on $\hat{b}_1$ should be $-A_0/\hat{\pi}_0$. The two versions differ by $(1-A_0/\hat{\pi}_0)\,\hat{b}_1$, which does not average to 0 in general, so only the corrected form supports the three robustness arguments. The book's third rearrangement (used for the Molina et al. argument) agrees with the corrected form.

ℹ Fine Point 21.2: Representations of the g-formula

The same g-formula can be written as standardization (the *joint density modeling* estimator of Section 21.1), as iterated conditional expectations (the ICE estimator, the basis of this section), or as an IP weighted mean (Section 21.2). Without parametric restrictions the three coincide, but in practice each suggests a different estimator (Hernán and Robins 2020, 291).

ℹ Technical Point 21.6: A Multiply Robust Estimator

This technical point gives a multiply robust plug-in estimator of $\mathrm{E}[Y^g]$ for static, dynamic, or random strategies g, based on Rotnitzky et al. (2017); without the weight covariate, the same algorithm is the ICE estimator of the g-formula. The book expects such estimators, fit with machine learning and sample splitting, to become more common as software matures, especially for failure-time outcomes.

4 21.4 G-Estimation for Time-Varying Treatments (pp. 289-296)

Definition 4.1 (Structural Nested Mean Model)**.** A **structural nested mean model** (SNMM) for a time-varying treatment specifies, for each $k=0,\ldots,K$,

$$\mathrm{E}\left[Y^{\bar{a}_{k-1},a_k,\underline{0}_{k+1}}-Y^{\bar{a}_{k-1},\underline{0}_k}\mid \bar{L}_k^{\bar{a}_{k-1}}=\bar{l}_k,\bar{A}_{k-1}=\bar{a}_{k-1},A_k=a_k\right]=a_k\,\gamma_k(\bar{a}_{k-1},\bar{l}_k,\beta),$$

where $(\bar{a}_{k-1},a_k,\underline{0}_{k+1})$ follows $\bar{a}_{k-1}$ through $k-1$, gives a_k at k, and gives no treatment from $k+1$ to K; $\gamma_k(\bar{a}_{k-1},\bar{l}_k,\psi^\dagger)$ is a known function of past treatment, covariate history, and a parameter value $\psi^\dagger$, with $\gamma_k(\bar{a}_{k-1},\bar{l}_k,0)=0$; and β is the unknown true value of that parameter.

Remark 4.1 (What an SNMM models). An SNMM models the effect on the mean of Y of a **last blip** of treatment a_k at time k, as a function of past treatment and covariate history. Because $\gamma_k(\cdot, 0) = 0$, the null hypothesis of no effect corresponds to $\beta = 0$.

Example 4.1 (A saturated SNMM for two time points). An SNMM (Definition 4.1) has **one equation per time point**. For Table 1 ($K = 1$), a saturated additive SNMM is

$$k = 0: \quad \mathrm{E}[Y^{a_0,a_1=0} - Y^{a_0=0,a_1=0} \mid A_0 = a_0] = \beta_0 a_0,$$
$$k = 1: \quad \mathrm{E}[Y^{a_0,a_1} - Y^{a_0,a_1=0} \mid L_1^{a_0} = l_1, A_0 = a_0, A_1^{a_0} = a_1] = a_1(\beta_{11} + \beta_{12} l_1 + \beta_{13} a_0 + \beta_{14} a_0 l_1).$$

The first equation is the effect of treating at time 0 and never again; the second, of treating at time 1 and never again. Under sequential exchangeability, $A_0 = a_0$ and $A_1^{a_0} = a_1$ can be dropped from the conditioning events.

Both equations are saturated: β_0 covers the single possible history at time 0, and the four β_{1j} cover the four histories (A_0, L_1) at time 1. The effect of a_1 is

- β_{11} when $A_0 = 0, L_1^{a_0} = 0$;
- $\beta_{11} + \beta_{12}$ when $A_0 = 0, L_1^{a_0} = 1$;
- $\beta_{11} + \beta_{13}$ when $A_0 = 1, L_1^{a_0} = 0$;
- $\beta_{11} + \beta_{12} + \beta_{13} + \beta_{14}$ when $A_0 = 1, L_1^{a_0} = 1$.

The two equations nested within each other are why the model is called *nested* (Hernán and Robins 2020, 291).

Definition 4.2 (Additive rank-preserving model for two time points). As in Chapter 14, g-estimation starts from the additive **rank-preserving** model for each individual i:

$$Y_i^{a_0,0} = Y_i^{0,0} + \psi_0 a_0,$$
$$Y_i^{a_0,a_1} = Y_i^{a_0,0} + \psi_{11} a_1 + \psi_{12} a_1 L_{1,i}^{a_0} + \psi_{13} a_1 a_0 + \psi_{14} a_1 a_0 L_{1,i}^{a_0}.$$

The second equation is only *locally* rank-preserving: ranks are preserved among individuals with the same a_0 and l_1.

⚠ Rank Preservation Is Implausible but Not Required

Rank preservation is biologically implausible, but it is not needed. G-estimation (the Chapter 14 method, extended here to time-varying treatment) is consistent for the SNMM parameters β of Definition 4.1 even if the rank-preserving model is wrong, provided the SNMM is correctly specified, sequential exchangeability for Y and consistency hold, and the model for treatment given past treatment and covariates is correctly specified.

The proof is in Robins (1994). To fit an unsaturated SNMM by g-estimation, positivity is not required (Hernán and Robins 2020, 292).

Definition 4.3 (Candidate counterfactuals for two time points). By consistency ($Y^{A_0,A_1} = Y$ and $L_1^{A_0} = L_1$), the model can be written in terms of observed data. Replacing ψ by a candidate value $\psi^\dagger$ defines the **candidate counterfactuals**

$$H_1(\psi^\dagger) = Y - \left(\psi_{11}^\dagger A_1 + \psi_{12}^\dagger A_1 L_1 + \psi_{13}^\dagger A_1 A_0 + \psi_{14}^\dagger A_1 A_0 L_1\right),$$
$$H_0(\psi^\dagger) = H_1(\psi^\dagger) - \psi_0^\dagger A_0.$$

Remark 4.2 (Candidate counterfactuals at the true parameter). When $\psi^\dagger = \psi$, $H_k(\psi^\dagger)$ equals the counterfactual $Y^{\bar{A}_{k-1},\bar{0}_k}$: observed treatment through $k-1$ and no treatment afterwards. Sequential exchangeability then says $H_k(\psi)$ is independent of A_k given the past.

Example 4.2 (G-Estimation in the Sequentially Randomized Experiment (Fine Point 21.3))**. Time 1.** Within each stratum of (A_0, L_1), the mean of $H_1(\psi)$ must not depend on A_1. The means of $H_1(\psi)$ by row of Table 1 are 84 and $84-\psi_{11}$ in stratum $(0,0)$, 52 and $52-\psi_{11}-\psi_{12}$ in $(0,1)$, 76 and $76-\psi_{11}-\psi_{13}$ in $(1,0)$, and 44 and $44-\psi_{11}-\psi_{12}-\psi_{13}-\psi_{14}$ in $(1,1)$ (for $A_1=0$ and $A_1=1$ respectively). Equating within strata:

$$\begin{aligned} 84 = 84-\psi_{11} &\implies \psi_{11}=0,\\ 52 = 52-\psi_{11}-\psi_{12} = 52-\psi_{12} &\implies \psi_{12}=0,\\ 76 = 76-\psi_{11}-\psi_{13} = 76-\psi_{13} &\implies \psi_{13}=0,\\ 44 = 44-\psi_{11}-\psi_{12}-\psi_{13}-\psi_{14} = 44-\psi_{14} &\implies \psi_{14}=0. \end{aligned}$$

Time 0. With all $\psi_{1j}=0$, $H_1(\psi)=Y$ and $H_0(\psi)=Y-\psi_0 A_0$. Exchangeability $Y^{0,0} \perp\!\!\!\perp A_0$ requires equal means of $H_0(\psi)$ in the two arms of 16,000 each:

$$\begin{aligned} A_0=0: &\quad 84\times 0.25 + 52\times 0.75 = 60,\\ A_0=1: &\quad (76-\psi_0)\times 0.5 + (44-\psi_0)\times 0.5 = 60-\psi_0, \end{aligned}$$

so $60 = 60-\psi_0$ and $\psi_0 = 0$. All parameters are 0, so $\mathrm{E}[Y^g] = 60$ under every static or dynamic strategy g, in agreement with the g-formula and IP weighting (Hernán and Robins 2020, 292–93).

4.1 General Structural Nested Mean Models

Definition 4.4 (Candidate counterfactuals for a general SNMM)**.** The candidate counterfactuals generalize to

$$H_k(\psi^\dagger) = Y - \sum_{j=k}^{K} A_j\, \gamma_j(\bar{A}_{j-1}, \bar{L}_j, \psi^\dagger).$$

Proposition 4.1 (Mean of the candidate counterfactuals at the true parameter)**.** *Suppose the SNMM of Definition 4.1 is correctly specified and consistency holds. Even when local rank preservation is false (as it essentially always is when treatment has an effect),* $\mathrm{E}\left[H_k(\beta) \mid \bar{A}_k, \bar{L}_k\right] = \mathrm{E}\left[Y^{\bar{A}_{k-1},\underline{0}_k} \mid \bar{A}_k, \bar{L}_k\right]$ *and* $\mathrm{E}[H_0(\beta)] = \mathrm{E}\left[Y^{\bar{0}}\right]$*. Hence, if* $\hat{\beta}$ *is consistent for* β*,* $\mathrm{E}\left[Y^{\bar{0}}\right]$ *is consistently estimated by the sample average of* $H_0(\hat{\beta})$ *(Hernán and Robins 2020, 294).*

Example 4.3 (Unsaturated blip functions)**.** Unsaturated examples of the blip function: β_1 (same effect for all histories and times), $\beta_1+\beta_2 k$ (effect varies linearly with time), and $\beta_1+\beta_2 k+\beta_3 a_{k-1}+\beta_4 l_k+\beta_5 l_k a_{k-1}$ (modification by the most recent treatment and covariate).

i Technical Point 21.7: Marginal Structural Models and Structural Nested Models

An SNMM is a semiparametric MSM if and only if γ_k does not depend on $\bar{l}_k$; the implied MSM is $\mathrm{E}[Y^{\bar{a}}] = \alpha_0 + \sum_{k=0}^{K} a_k \gamma_k(\bar{a}_{k-1}, \beta)$ with $\alpha_0 = \mathrm{E}\left[Y^{\bar{0}}\right]$. Such an SNMM additionally assumes no effect modification by past covariates. If that assumption holds, g-estimation is more efficient than IP weighting; if it fails, g-estimation is biased while IP weighting of the MSM is not: a variance-bias trade-off.

4.2 G-Estimation Algorithm (One Parameter)

Algorithm 4.1 (G-estimation of a one-parameter SNMM). *Take* $\gamma_k(\bar{a}_{k-1}, \bar{l}_k, \psi^\dagger) = \psi^\dagger$ *(true value* β_1*), so* $H_k(\psi^\dagger) = Y - \sum_{j=k}^{K} A_j \psi^\dagger$.

1. *For each* $\psi^\dagger$ *on a grid from* ψ_{low} *to* ψ_{up} *(e.g., steps of 0.1), compute* $H_k(\psi^\dagger)$ *for every person and time.*
2. *Fit the pooled logistic model* $\operatorname{logit}\Pr[A_k = 1 \mid H_k(\psi^\dagger), \bar{L}_k, \bar{A}_{k-1}] = \alpha_0 + \alpha_1 H_k(\psi^\dagger) + \alpha_2 W_k$, *where* $W_k = w_k(\bar{L}_k, \bar{A}_{k-1})$ *and each person contributes* $K+1$ *rows.*
3. *The g-estimate* $\hat{\beta}$ *is the* $\psi^\dagger$ *with* $\hat{\alpha}_1$ *closest to 0; the 95% confidence interval is the set of* $\psi^\dagger$ *with* $P > 0.05$ *for* $\alpha_1 = 0$.

Remark 4.3 (G-estimation as solving an estimating equation). Equivalently, $\hat{\beta}$ is the $\psi^\dagger$ at which the score test of $\alpha_1 = 0$ has $P = 1$, that is, the solution of

$$\sum_{i=1}^{N} \sum_{k=0}^{K} \{A_{i,k} - \operatorname{expit}(\hat{\alpha}_0 + \hat{\alpha}_2 W_{i,k})\} H_{i,k}(\psi^\dagger) = 0,$$

with $\hat{\alpha}_0, \hat{\alpha}_2$ from the model fit with $\alpha_1 = 0$.

Proposition 4.2 (Consistency of the g-estimator). *Let* $\hat{\beta}$ *be the solution of the estimating equation of Remark 4.3 (equivalently, the limit of the grid-search estimate of alg. 4.1 as the grid step goes to 0). Then* $\hat{\beta}$ *is consistent for the true* β_1 *if (i) the one-parameter SNMM of alg. 4.1,* $\gamma_k(\bar{a}_{k-1}, \bar{l}_k, \beta) = \beta_1$, *is correct, (ii) sequential exchangeability for* Y *holds, (iii) consistency holds,* $Y^{\bar{A}} = Y$, *(iv) the treatment model* $\operatorname{logit}\Pr[A_k = 1 \mid \bar{L}_k, \bar{A}_{k-1}] = \alpha_0 + \alpha_2 W_k$ *is correct, and (v)* $H_k(\psi^\dagger)$ *enters the treatment model linearly (Technical Point 14.2).*

Remark 4.4 (G-estimation of a vector parameter). **Vector** β: for the five-parameter blip $\beta_0 + \beta_1 k + \beta_2 a_{k-1} + \beta_3 l_k + \beta_4 l_k a_{k-1}$, the treatment model needs five terms involving $H_k(\psi^\dagger)$, e.g. $\alpha_0 + H_k(\psi^\dagger)(\alpha_1 + \alpha_2 k + \alpha_3 A_{k-1} + \alpha_4 L_k + \alpha_5 L_k A_{k-1}) + \alpha_6 W_k$, and $\hat{\beta}$ solves the corresponding five-dimensional estimating equation. The choice of these terms affects the width of the confidence interval, not consistency. A grid search is impractical beyond about two dimensions (20^5 points for 20 values per component), but for an SNMM linear in β the estimating equation has a closed-form solution (Technical Point 21.8, which also gives a 2^{K+1} multiply robust version) (Hernán and Robins 2020, 295–96).

4.3 From $\hat{\beta}$ to Counterfactual Means

Remark 4.5 (Counterfactual means after g-estimation). Given $\hat{\beta}$, how to estimate a counterfactual mean depends on the strategy and the blip function:

- $\mathrm{E}\left[Y^{\bar{0}}\right]$: the sample average of $H_0(\hat{\beta})$.

- If there is no effect modification by past covariates, $\gamma_k(\bar{a}_{k-1}, \bar{l}_k, \beta) = \gamma_k(\bar{a}_{k-1}, \beta)$, and for a static $\bar{a}$

$$\hat{\mathrm{E}}[Y^{\bar{a}}] = \hat{\mathrm{E}}\left[Y^{\bar{0}}\right] + \sum_{k=0}^{K} a_k \gamma_k(\bar{a}_{k-1}, \hat{\beta}).$$

- If γ_k depends on $\bar{l}_k$, or g is dynamic, simulate L_k from a model for $f(l_k \mid \bar{a}_{k-1}, \bar{l}_{k-1})$ (Technical Point 21.9).

i Technical Point 21.9: Estimating a Counterfactual Mean After g-estimation of a Structural Nested Mean Model

Estimate $\mathrm{E}\left[Y^{\bar{0}}\right]$ by the average of $H_0(\hat{\beta})$; fit a model for $f(l_k \mid \bar{a}_{k-1}, \bar{l}_{k-1})$; simulate covariate histories under g; and add to $\hat{\mathrm{E}}\left[Y^{\bar{0}}\right]$ the average over simulations of $\sum_{j=0}^{K} a_{v,j} \gamma_j(\bar{a}_{v,j-1}, \bar{l}_{v,j}, \hat{\beta})$. The result is consistent for $\mathrm{E}[Y^g]$ if the covariate model and the SNMM are correct and either the treatment model or the outcome model $\mathrm{E}\left[Y^{\bar{A}_{k-1}, \underline{0}_k} \mid \bar{L}_k, \bar{A}_{k-1}\right]$ is correct. Under the null, $\gamma_j(\cdot, \hat{\beta})$ converges to 0 whenever $\hat{\beta}$ is consistent for $\beta = 0$, even when the covariate model is wrong. So, under the identifiability conditions, this procedure **preserves the null**, unlike the parametric g-formula, provided $\Pr[A_k = 1 \mid \bar{L}_k, \bar{A}_{k-1}]$ is known (as in a sequentially randomized experiment) or either the treatment model or the outcome model is correct at each k (Hernán and Robins 2020, 297).

i Technical Point 21.13: Formal Definition of a General Structural Nested Mean Model

This technical point generalizes SNMMs to blips followed by an arbitrary strategy g rather than "never treat" (Robins 2004), and notes identification under a time-varying parallel-trends assumption instead of sequential exchangeability (Hernán and Robins 2020, 303).

5 21.5 Censoring Is a Time-Varying Treatment (pp. 297-299)

Part II treated censoring C as time-fixed.

Definition 5.1 (Time-varying censoring)**.** More realistically, censoring is time-varying: $C_1, C_2, \ldots, C_{K+1}$, where $C_m = 0$ if the individual is still uncensored at m and 1 otherwise.

- Censoring is **monotone**: $C_m = 0$ implies $C_1 = \cdots = C_{m-1} = 0$.
- $C_0 = 0$ for everybody, since otherwise they would not be in the study.
- After censoring, treatments, confounders, and outcomes are unobserved, so analyses are restricted to uncensored person-times.

Definition 5.2 (The g-formula with time-varying censoring)**.** With time-varying censoring, the g-formula of Definition 1.1 for strategy $\bar{a}$ becomes

$$\sum_{\bar{l}} \mathrm{E}[Y \mid \bar{C} = \bar{0}, \bar{A} = \bar{a}, \bar{L} = \bar{l}] \prod_{k=0}^{K} f(l_k \mid \bar{c}_k = \bar{0}, \bar{a}_{k-1}, \bar{l}_{k-1}).$$

Proposition 5.1 (The censored g-formula identifies the uncensored counterfactual mean)**.** *If sequential exchangeability, positivity, and consistency hold with A_m replaced by the joint treatment (A_m, C_{m+1}) at every m, then the g-formula of Definition 5.2 equals $\mathrm{E}\left[Y^{\bar{a}, \bar{c}=\bar{0}}\right]$: the mean outcome had everyone followed $\bar{a}$ and nobody been lost to follow-up.*

The superscript $\bar{c} = \bar{0}$ makes explicit the contrast many investigators have in mind when they speak of "the effect of treatment" (Hernán and Robins 2020, 298).

> ⚠ Conditioning on Being Uncensored Can Create Selection Bias
>
> Conditioning on being uncensored induces selection bias under the null when C is a collider on a path between A and Y, or a descendant of one (Chapter 8).

5.1 IP Weighting with Censoring

Algorithm 5.1 (IP weighting with time-varying censoring)**.** *Fit, for example,* $\mathrm{E}\left[Y \mid \bar{A}, \bar{C} = \bar{0}\right] = \theta_0 + \theta_1 \mathrm{cum}(\bar{A})$ *in the pseudo-population created by* $W^{\bar{A}} \times W^{\bar{C}}$ *or* $SW^{\bar{A}} \times SW^{\bar{C}}$*, where*

$$W^{\bar{C}} = \prod_{k=1}^{K+1} \frac{1}{\Pr\left[C_k = 0 \mid C_{k-1} = 0, \bar{A}_{k-1}, \bar{L}_{k-1}\right]}, \qquad SW^{\bar{C}} = \prod_{k=1}^{K+1} \frac{\Pr\left[C_k = 0 \mid C_{k-1} = 0, \bar{A}_{k-1}\right]}{\Pr\left[C_k = 0 \mid C_{k-1} = 0, \bar{A}_{k-1}, \bar{L}_{k-1}\right]}.$$

Numerator and denominator are estimated with separate logistic models.

Remark 5.1 (The pseudo-population created by censoring weights).

- **Nonstabilized** weights replace censored individuals by copies of uncensored ones with the same treatment and covariate history: the pseudo-population has the size of the study population **before** censoring, and censoring is abolished.
- **Stabilized** weights keep the size **after** censoring (the same proportion censored at each k), but make censoring occur at random with respect to $\bar{L}_k$: selection, but no selection bias.
- With nonstabilized weights the pseudo-population has no arrows from L_k or A_k into later C_m $(m > k)$. With stabilized weights it has no arrows from L_k into later C_m, but censoring may still depend on past treatment, because the numerator conditions on $\bar{A}_{k-1}$ (as for SW^C in Chapter 12). Either way, censoring no longer depends on the measured covariate history, so IP weighting can estimate the joint effect of $(\bar{A}, \bar{C})$ even when $\bar{L}$ is affected by prior treatment.

Remark 5.2 (G-estimation with censoring). **G-estimation with censoring**: first create, with nonstabilized weights $W^{\bar{C}}$, a pseudo-population in which nobody is censored, then g-estimate in it.

> 💡 Check the Mean of the Stabilized Weights
>
> After fitting the censoring models, compute the sample mean of the estimated $SW^{\bar{C}}$. Correctly specified models give a mean close to 1 (Hernán and Robins 2020, 298), so for $SW^{\bar{C}}$ a mean well away from 1 suggests that the numerator or the denominator model is misspecified. The same check applies to $SW^{\bar{A}}$ with a narrower reading: given a correct denominator f_k, taking iterated expectations over a_k at each time k gives $\mathrm{E}\left[\prod_k n_k / f_k\right] = 1$ for any proper conditional numerator n_k, so for $SW^{\bar{A}}$ a mean away from 1 points to the denominator (treatment) model, or to extreme weights from near-violations of positivity, not to the numerator.

⚠ Fact-Check: Stabilized Weights Keep Arrows from Past Treatment

Fact-check note: the book states that "regardless of the type of IP weights used", the pseudo-population has no arrows from L_k and A_k into future C_m (Hernán and Robins 2020, 299). For stabilized weights this holds for L_k only: their numerator $\Pr[C_k = 0 \mid C_{k-1} = 0, \bar{A}_{k-1}]$ keeps the dependence of censoring on past treatment, which is harmless because the analysis conditions on $\bar{A}$.

ℹ Technical Point 21.10: Survival Analysis with Time-Varying Treatments

To estimate the risk $\Pr[D_{k+1}^{\bar{a},\bar{c}=\bar{0}} = 1]$, either use the g-formula with models for the discrete-time hazards and for the confounder densities, conditional on being event-free and uncensored (e.g., pooled logistic models, as in Chapter 17), or fit a pooled logistic hazards model in which each person-time receives the time-varying weight $W_k^{\bar{A}} \times W_k^{\bar{C}}$ (or its stabilized version); the latter estimates a marginal structural pooled logistic model (Robins 1998) (Hernán and Robins 2020, 299).

6 21.6 The Big G-Formula (pp. 300-303)

This part of the book relies on sequential exchangeability given the measured L and identification by the g-formula, because few realistic longitudinal analyses use other identifying conditions (such as the front door criterion). But all identifying formulas are mathematically linked to the g-formula based on **all** variables, measured and unmeasured.

Definition 6.1 (The Big G-Formula)**.** Given a causal DAG with treatment A, outcome Y, measured covariates L, and unmeasured variables U, let $X = (L, U)$. Every parent of a treatment node is in A or X, so X ensures sequential exchangeability. The **big g-formula** is the g-formula with L replaced by X. If positivity held, it would identify $\mathrm{E}[Y^g]$ under any strategy g (Hernán and Robins 2020, 300).

Definition 6.2 (Factuals)**.** *Factuals* are variables of the actual world, in contrast to counterfactuals; some factuals, like U, are unavailable for analysis because they were not measured (Hernán and Robins 2020, 300).

Remark 6.1 (The big g-formula cannot be computed directly). The big g-formula is a function of the distribution of the factuals (A, L, Y, U) (Definition 6.2) alone, but it includes unmeasured U, so it cannot be computed directly.

6.1 Reducing the Big G-Formula to Observed Data

Remark 6.2 (When the big g-formula reduces to an observed-data formula). The key question: can the big g-formula be rewritten as a functional of the distribution of the observed (A, L, Y) only? If so, the new formula reproduces the big g-formula and can be used in data analysis.

- Under the **front door** criterion (Figure 7.14), the big g-formula for $\mathrm{E}[Y^a]$ reduces to the front door formula (Technical Point 21.11).
- In general, whether such a reduction exists, and what the formula is, using only the d-separations implied by the DAG, was settled by Tian and Pearl (2002), Shpitser and Pearl (2006), and Huang and Valtorta (2006).

⚠ A Reduction Is Causal Only If the DAG Is

These are purely mathematical questions about distributions obeying d-separation; they acquire a causal meaning only if the DAG is truly causal, which we can never know for certain in an observational study.

Example 6.1 (The Front Door Formula from the Big G-Formula (Technical Point 21.11))**.** Under Figure 7.14 ($A \to M \to Y$, with U a common cause of A and Y), the big g-formula for $\Pr[Y^a = y]$ is

$$\sum_m \sum_u \Pr[Y = y \mid M = m, A = a, U = u] \Pr[M = m \mid A = a, U = u] \Pr[U = u].$$

Step 1: by $A \perp\!\!\!\perp Y \mid M, U$, drop $A = a$ from the first factor; by $U \perp\!\!\!\perp M \mid A$, replace $\Pr[M = m \mid A = a, U = u]$ by $\Pr[M = m \mid A = a]$; and write $\Pr[U = u] = \sum_{a'} \Pr[U = u \mid A = a'] \Pr[A = a']$:

$$\sum_m \Pr[M = m \mid A = a] \sum_u \Pr[Y = y \mid M = m, U = u] \sum_{a'} \Pr[U = u \mid A = a'] \Pr[A = a'].$$

Step 2: by $U \perp\!\!\!\perp M \mid A$, $\Pr[U = u \mid A = a'] = \Pr[U = u \mid M = m, A = a']$, and by $A \perp\!\!\!\perp Y \mid M, U$, $\Pr[Y = y \mid M = m, U = u] = \Pr[Y = y \mid M = m, A = a', U = u]$:

$$\sum_m \Pr[M = m \mid A = a] \sum_{a'} \left\{ \sum_u \Pr[Y = y \mid M = m, A = a', U = u] \Pr[U = u \mid M = m, A = a'] \right\} \Pr[A = a'].$$

Step 3: the braces sum over u to $\Pr[Y = y \mid M = m, A = a']$, giving the front door formula

$$\Pr[Y^a = y] = \sum_m \Pr[M = m \mid A = a] \sum_{a'} \Pr[Y = y \mid M = m, A = a'] \Pr[A = a'],$$

which involves observed variables only. This proof does not require that counterfactuals Y^m exist (Hernán and Robins 2020, 301).

Remark 6.3 (Other proofs of the front door formula). Technical Point 21.11 also gives a coupling argument, and Technical Point 21.12 a proof based on a SWIG property (Shpitser et al. 2022): if a fixed treatment node a_m is d-separated from B^a given C^a on the SWIG, then $\Pr[B^a = b \mid C^a = c]$ does not depend on a_m (Hernán and Robins 2020, 301–2).

7 Summary

- In Table 1, the treatment has no effect; the g-formula, IP weighting, and g-estimation all give $\mathrm{E}[Y^{1,1}] - \mathrm{E}[Y^{0,0}] = 60 - 60 = 0$, where traditional methods failed.
- **G-formula**: standardize to the distribution of each L_k given past treatment and covariates; estimate with parametric models (the parametric g-formula), which is subject to the g-null paradox.
- **IP weighting**: $W^{\bar{A}}$ or $SW^{\bar{A}}$, a product over time, fit marginal structural models by weighted regression; $SW^{\bar{A}}$ preferred for unsaturated models.
- **Doubly robust**: sequential outcome regressions with the time-varying IP weight as a covariate, consistent if either set of models is correct, and in fact $K + 2$ robust.
- **G-estimation**: one SNMM equation per time; find $\psi^\dagger$ making $H_k(\psi^\dagger)$ independent of A_k given the past.

- **Censoring** is handled like treatment: the joint strategy $(\bar{a}, \bar{c} = \bar{0})$ and weights $W^{\bar{C}}$ or $SW^{\bar{C}}$.
- The **big g-formula** uses all variables, measured and unmeasured; other identifying formulas, such as the front door formula, are reductions of it to observed data.

8 References

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. https://miguelhernan.org/whatifbook.